



ME 210: Fluid Mechanics

Course Code	ME 210
Course Name	Fluid Mechanics
L-T-P-C	2.5-0.5-0-3
Prerequisites	None
Intended for	B.Tech.
Distribution	Core
One-time approval	5th Senate (Oct 2013)

Course Contents

- **Introduction**
definition of fluid, liquids and gases, continuum hypothesis, compressible and incompressible fluid/flow, viscosity, stress field, Newtonian and non-Newtonian fluids [6 Lectures]
- **Fluid Statics**
Pascal's law, hydrostatic pressure, standard atmosphere, manometry, center of pressure, forces on partially and fully submerged bodies, buoyancy, metacentric height, stability, rigid body motion. [6 Lectures]
- **Fluid Kinematics**
Lagrangian and Eulerian description of fluid motion, elementary flows, vorticity and circulation, flow lines, stream lines, stream functions, rotational and ir-rotational flows, flow visualization. [6 Lectures]
- **Fluid Dynamics**
Newton's second law, fundamental equations of mass, momentum and energy, Reynolds transport theorem, Integral formulation of governing equations, differential formulation, Euler's equation, Bernoulli's equation, Navier-Stokes equation. [6 Lectures]
- **Internal Flows**
fully developed flow, Couette Flow, Hagen-Poiseuille flow, flow through ducts, channels, Venturi, Orifice, flow measurements, friction factor and head loss calculations, Moody's chart, open-channel flow. [8 Lectures]
- **Dimensional Analysis**
scaling and similarity, similitude and dimensional analysis, Buckingham π – theorem, non-dimensional parameters, model testing. [4 Lectures]
- **External Flows**
Boundary layer flows, flow over an aerofoil, flow over a cylinder and sphere, laminar and turbulent flows, flow separation, lift and drag, D'Alembert paradox, von Karman integral equation, displacement and momentum thickness. [4 Lectures]

Text Books

1. Fox and Mc Donald, **Introduction to Fluid Mechanics**, 7th Edition, John Wiley, 2009.
2. White FM,, **Fluid Mechanics**, 6th Edition, Tata McGraw Hill, 2007.

References

1. Yuan SW, **Foundations of Fluid Mechanics**, 2nd Edition, Printice Hall, 1988.
2. Streeter VL, Wylie EB and Bedford KW, **Fluid Mechanics**, 9th Edition, McGraw Hill, 1998.